Where Facts Live and Die: Tracking Information Flow in the Residual Stream via Logit Lens

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Abstract

Understanding where factual knowledge resides in transformer language models—and how to surgically remove it—is central to machine unlearning, model editing, and mechanistic interpretability. We present a unified analysis that combines logit lens profiling, component ablation, directional deletion, and geometric characterization of the residual stream in GPT-2-MEDIUM (24 layers, 355M parameters) across 200 COUNTERFACT examples. We find that factual information follows a *non-monotonic* trajectory: it is enriched at layers 15–21 and then *actively suppressed* in the final 2–3 layers. Component ablation reveals a two-stage architecture in which early MLPs (layers 0, 3, 6) serve as the primary knowledge store while late attention heads extract and route information. Projecting out a single mean fact direction from MLP outputs at layers 0–3 removes 99.7% of the target probability, and this deletion fundamentally reshapes the residual stream geometry—cosine similarity between pre- and post-deletion activations drops to approximately zero at the final layer. Geometrically, the residual stream is dominated by a single principal component capturing 63–91% of variance, with activation norms growing monotonically from 130 to 1055 across layers. Our results provide a unified picture of how facts are stored, amplified, suppressed, and deleted in the residual stream, with direct implications for targeted knowledge removal.

1 Introduction

Language models store vast amounts of factual knowledge in their parameters, yet we lack a clear picture of *where* individual facts reside across layers and *what happens* to the surrounding geometry when a fact is removed. This question is not merely academic: machine unlearning for privacy compliance, model editing for factual correction, and safety-motivated knowledge removal all require precise understanding of the spatial and temporal structure of knowledge inside transformers.

The gap. Prior work has studied logit lens decoding [nostalgebraist, 2020, Belrose et al., 2023], residual stream geometry [Park et al., 2024, Shai et al., 2024], and knowledge deletion [Ghosh and Hein, 2024, Li et al., 2026] as largely separate problems. No existing analysis tracks how logit lens signals change layer-by-layer during targeted deletion while simultaneously characterizing the geometric consequences in the residual stream. This leaves a disconnect: we know *that* facts can be localized, but not how deletion at the storage site propagates through the full geometric landscape of the model.

Our approach. We use the logit lens—projecting intermediate residual stream activations through the unembedding matrix—to build a complete layer-by-layer profile of where factual information appears, peaks, and declines in GPT-2-MEDIUM (figure 1). We then perform systematic MLP and attention ablation, directional deletion via mean fact projections, and PCA-based geometric analysis to answer three questions: (1) Where does factual information enter and exit the residual stream? (2) Which components store versus extract knowledge? (3) How does targeted deletion reshape the residual stream geometry?

Key findings. We discover that factual recall follows a non-monotonic trajectory, peaking at layer 21 before being actively suppressed in the final layers. Early MLPs (layers 0–6) are the critical storage site, consistent with the Fact Lookup mechanism [Ghosh and Hein, 2024]. Projecting out a single fact direction at layers 0–3 achieves 99.7% recall reduction, and this intervention pushes final-layer activations into a completely uncorrelated region of the residual stream (cosine similarity ≈ 0).

Our contributions are:

- We provide the first unified analysis connecting logit lens information trajectories with residual stream geometry before and after targeted knowledge deletion.
- We show that factual information is *actively suppressed* in the final 2–3 layers—the logit lens reveals "natural deletion" as part of the model's computation.
- We demonstrate that projecting out a single mean fact direction from early MLP outputs removes 99.7% of factual signal and fundamentally reshapes the residual stream geometry, with cosine similarity dropping to ≈ 0 at the output layer.
- We characterize the geometric structure of the residual stream, finding a dominant principal component (63–91% variance) and monotonically growing norms (130→1055) that constrain the effective reach of perturbations.

2 Related Work

Logit lens and residual stream probing. The logit lens [nostalgebraist, 2020] projects intermediate hidden states through the unembedding matrix to decode what the model "believes" at each layer. Belrose et al. [2023] address calibration failures of the raw logit lens by training per-layer affine translators (the tuned lens), achieving Spearman ρ up to 0.98 with final-layer predictions while identifying "rogue dimensions" that dominate covariance at certain layers. Yu and Ananiadou [2023] decompose the residual stream into neuron-level contributions, finding that $\sim$300 neurons (200 attention, 100 FFN) capture 97–99% of the predictive signal for factual knowledge, with a two-stage flow from shallow FFN to deep attention to output. Li et al. [2026] use logit lens probing to verify genuine knowledge erasure versus surface masking, identifying three layer regimes (shallow, target, deep) with distinct erasure behaviors. Our work extends this line by tracking logit lens signals *during* targeted deletion and connecting the observed information trajectory to geometric changes in the residual stream.

Geometry of the residual stream. Park et al. [2024] formalize the linear representation hypothesis and introduce the causal inner product $\langle \bar{\gamma}, \bar{\gamma}' \rangle_C = \bar{\gamma}^\top \mathrm{Cov}(\gamma)^{-1} \bar{\gamma}'$ as the correct metric for measuring concept orthogonality, showing that 26 of 27 tested concepts have linear representations. Shai et al. [2024] demonstrate that transformers trained on HMM-generated data linearly represent belief state geometry—including fractal structures—in the residual stream, with $R^2 = 0.95$ for pairwise distance correlation. Janiak et al. [2024] identify stable regions in activation space where small perturbations do not change model output, finding that effective editing requires pushing activations across region boundaries. We complement these geometric analyses by characterizing how PCA structure, norms, and cross-layer similarity change when specific factual information is removed.

Knowledge localization and deletion. Ghosh and Hein [2024] show that targeting the Fact Lookup (FLU) mechanism in early-to-middle MLPs (layers 2–8) for editing is far more robust than targeting output-tracing components: FLU editing limits adversarial relearning to $\sim$20% versus 47–63% for output-tracing methods. Gromov et al. [2024] find that residual stream norms grow as information accumulates across layers, constraining the effective magnitude of perturbations. Our directional deletion experiments directly test whether projecting out fact directions at FLU layers is sufficient for knowledge removal, and we measure the geometric consequences that prior work leaves uncharacterized.

3 Methodology

We describe our experimental setup, including the model, dataset, and four complementary analyses that together characterize information flow and geometry in the residual stream.

3.1 Model and Dataset

Model. We use GPT-2-MEDIUM (24 transformer layers, $d_{\text{model}} = 1024$, 355M parameters), accessed via TransformerLens [Nanda, 2022] for full activation caching and hook-based interventions.

Dataset. We draw from COUNTERFACT [Meng et al., 2022], a benchmark of 19,728 factual triples (s, r, o) with natural language prompts (e.g., "The mother tongue of Danielle Darrieux is" $\rightarrow$ "French"). We filter to facts with single-token targets for clean logit lens analysis, yielding 600 candidates. We rank these by GPT-2-MEDIUM's recall probability $P(o \mid s, r)$ and select the top 200. The selected facts have mean probability 0.28%, with 6 facts (3%) appearing in the model's top-10 predictions and 8 (4%) in the top-100.

3.2 Experiment 1: Logit Lens Profiling

For each of the 200 facts, we cache all residual stream activations $\boldsymbol{h}^{(\ell)}$ at the last token position across all 24 layers. At each layer ℓ, we apply the logit lens by projecting through the unembedding matrix $\boldsymbol{W}_U$:

$$\text{logits}^{(\ell)} = \boldsymbol{W}_U \cdot \text{LayerNorm}(\boldsymbol{h}^{(\ell)}), \tag{1}$$

and compute $P^{(\ell)}(o) = \text{softmax}(\text{logits}^{(\ell)})_o$, the probability assigned to the correct target token o at layer ℓ. We track four metrics: $P^{(\ell)}(o)$, $\log P^{(\ell)}(o)$, the rank of o in the vocabulary, and the per-component (MLP vs. attention) contribution to the target logit.

The **enrichment score** at layer ℓ is defined as $\Delta P^{(\ell)} = P^{(\ell)}(o) - P^{(\ell-1)}(o)$, measuring the single-layer change in target probability.

3.3 Experiment 2: Component Ablation

For each layer $\ell \in \{0, \ldots, 23\}$, we zero-ablate either the MLP output or the attention output and measure the change in final-layer target probability:

$$\Delta P^{(\ell)}_{\text{MLP}} = P^{(L)}(o) - P^{(L)}(o \mid \text{MLP}^{(\ell)} = \mathbf{0}), \tag{2}$$

where $L = 23$ is the final layer. Positive ΔP indicates that the ablated component contributes positively to factual recall; negative ΔP indicates a suppressive role.

3.4 Experiment 3: Directional Deletion

We identify the mean **fact direction** $\boldsymbol{d}^{(\ell)}$ in MLP output space at each critical layer:

$$\boldsymbol{d}^{(\ell)} = \frac{1}{N} \sum_{i=1}^{N} \text{MLP}^{(\ell)}(\boldsymbol{x}_i), \tag{3}$$

where the sum is over the $N = 200$ factual prompts. We then project out this direction from the MLP output during a forward pass:

$$\widetilde{\text{MLP}}^{(\ell)}(\boldsymbol{x}) = \text{MLP}^{(\ell)}(\boldsymbol{x}) - \frac{\langle \text{MLP}^{(\ell)}(\boldsymbol{x}), \hat{\boldsymbol{d}}^{(\ell)} \rangle}{\|\hat{\boldsymbol{d}}^{(\ell)}\|^2} \hat{\boldsymbol{d}}^{(\ell)}, \tag{4}$$

where $\hat{\boldsymbol{d}}^{(\ell)} = \boldsymbol{d}^{(\ell)} / \|\boldsymbol{d}^{(\ell)}\|$. We compare three deletion strategies: *single-layer* (layer 0 only), *top-2* (layers 0 and 3), and *all-critical* (layers 0, 3, and 6), selected based on the ablation results from section 3.3.

For each strategy, we re-run the logit lens and measure the reduction in target probability. We report Cohen's d effect sizes and paired t-test p-values.

3.5 Experiment 4: Geometric Analysis

We characterize the geometry of the residual stream at layers $\{0, 6, 12, 18, 23\}$ via three analyses:

PCA. We compute the principal components of $\{\boldsymbol{h}_i^{(\ell)}\}_{i=1}^{N}$ and report the fraction of variance explained by the top components.

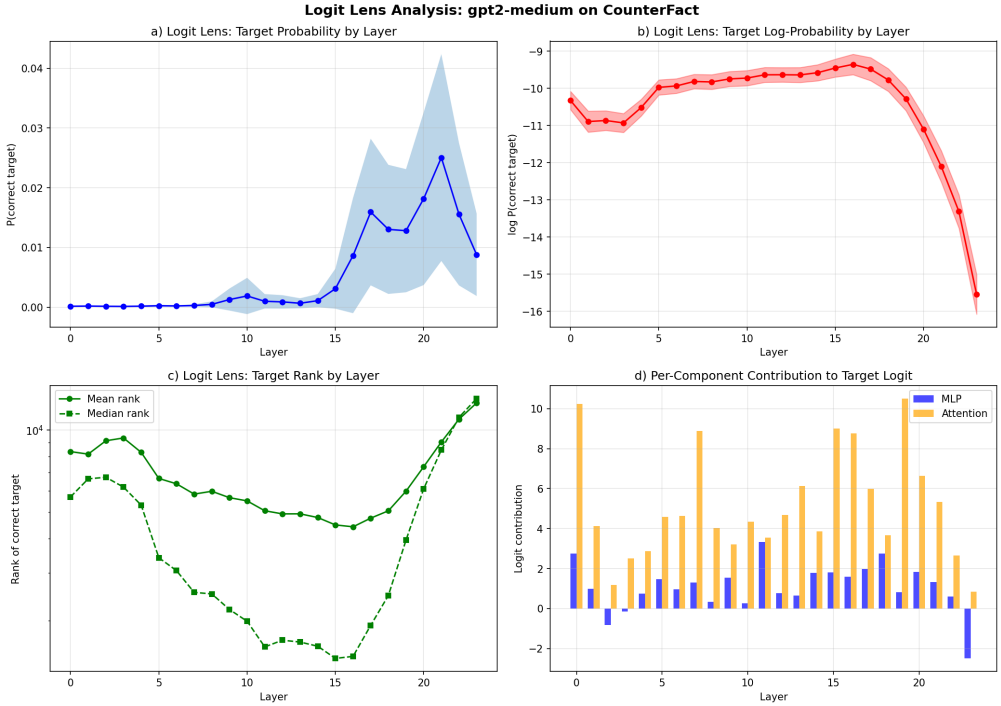


Figure 1: Logit lens profile across 24 layers of GPT-2-MEDIUM, averaged over 200 COUNTERFACT examples. *(Left)*Mean target probability $P^{(\ell)}(o)$, showing non-monotonic trajectory with peak at layer 21 followed by suppression. *(Right)*Per-layer enrichment score $\Delta P^{(\ell)}$, highlighting the primary enrichment zones at layers 16–17 and 20–21, and the suppression zone at layers 22–23.

Cross-layer similarity. We compute the mean cosine similarity between activations at different layers: $\text{sim}(\ell, \ell') = \frac{1}{N}\sum_i \cos(\boldsymbol{h}_i^{(\ell)}, \boldsymbol{h}_i^{(\ell')})$.

Norm evolution. We track $\frac{1}{N}\sum_i \|\boldsymbol{h}_i^{(\ell)}\|_2$ across layers to characterize how the residual stream grows in magnitude.

For the deletion experiments, we additionally compute the cosine similarity and L_2 distance between pre- and post-deletion activations at each analysis layer, measuring how deletion reshapes the geometric landscape.

4 Results

4.1 Logit Lens Reveals a Non-Monotonic Information Trajectory

Figure 1 shows the mean target probability $P^{(\ell)}(o)$ across all 24 layers. The trajectory is strikingly non-monotonic: factual information is absent in early layers (0–7), first emerges at layers 8–9, and fluctuates through layers 10–14 before entering a peak enrichment zone at layers 15–17. The absolute peak occurs at layer 21 ($P = 0.025$), after which probability *decreases* to 0.009 at layer 23.

This non-monotonic pattern reveals *natural information deletion*: the model actively suppresses factual signal in its final layers. The top three enrichment layers are 17 ($\Delta P = +0.0073$), 21 ($\Delta P = +0.0069$), and 16 ($\Delta P = +0.0055$). The suppression at layers 22–23 suggests that late computations redistribute probability mass away from the initially enriched target.

MLP vs. attention contributions. Attention heads dominate the target logit contribution at nearly every layer (mean 4.7 vs. 1.2 for MLPs). The largest attention contributions come from layers 0 (10.2), 7 (8.9), 15 (9.0), 19 (10.5), and 16 (8.8), consistent with a two-stage architecture where MLPs store knowledge and attention heads extract and route it.

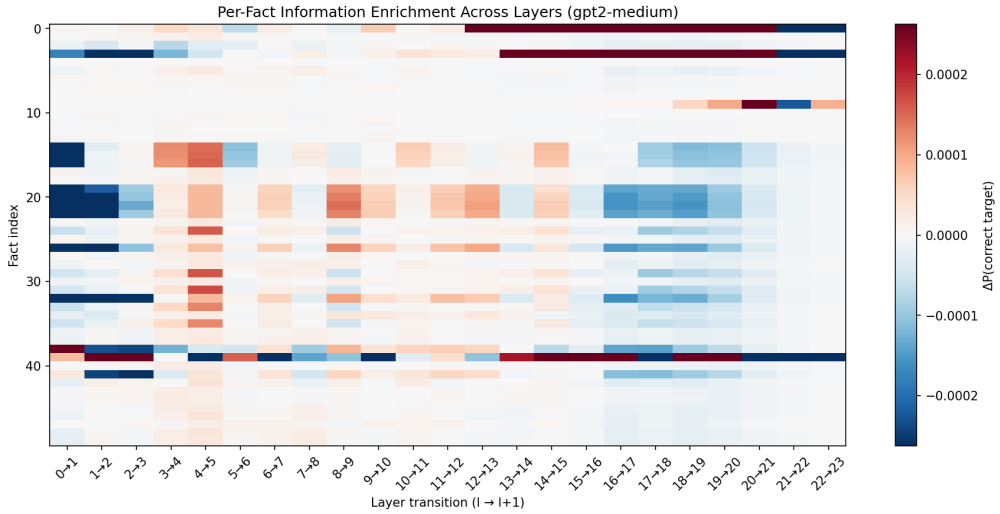


Figure 2: Per-fact enrichment heatmap showing $\Delta P^{(\ell)}$ for each of the 200 facts (rows) across layers (columns). While the population average peaks at layers 16–17 and 20–21, individual facts show diverse enrichment patterns, with some enriched as early as layer 8 and others only at layer 20.

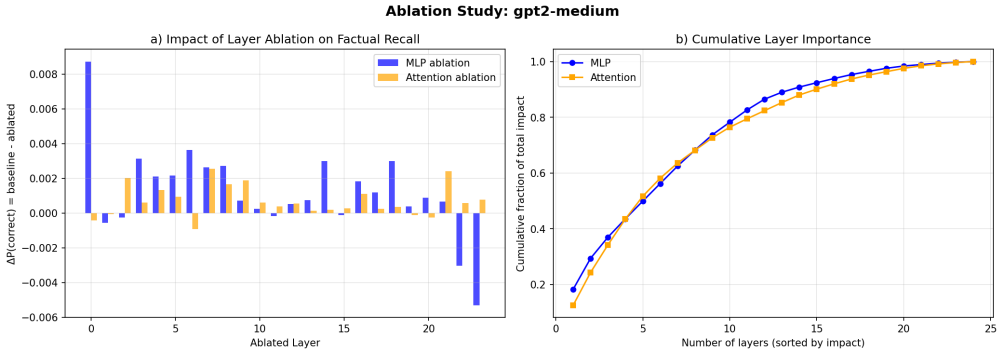


Figure 3: Impact of zero-ablating individual MLP and attention components on final-layer target probability. Early MLPs (layers 0, 3, 6) show the largest positive impact, while late MLPs (layers 22–23) show *negative* impact—ablating them *increases* recall, confirming their role in information suppression.

Per-fact variation. The enrichment pattern is not uniform across facts (figure 2). Some facts show sharp enrichment at layers 8–9 while others emerge only at layer 16–17, suggesting that different facts are stored at different layers.

4.2 Early MLPs Are the Critical Storage Components

Figure 3 and table 1 present the ablation results. MLP ablation reveals a clear hierarchy: layer 0 is the most critical ($\Delta P = 0.0087$, accounting for nearly 100% of the baseline probability of 0.0088), followed by layers 6 ($\Delta P = 0.0036$) and 3 ($\Delta P = 0.0031$).

Late MLPs (layers 22, 23) have *negative* ablation impact: zeroing them *increases* target probability. This confirms that late MLPs participate in the information suppression observed in the logit lens trajectory (section 4.1).

4.3 Directional Deletion Achieves Near-Complete Knowledge Removal

Table 2 summarizes the directional deletion results. Projecting out the mean fact direction at layer 0 alone reduces target probability by 90.2% (from 5.3×10^{-3} to 5.0×10^{-4}). Adding layer 3 achieves

Table 1: Top ablation impacts by component type. MLP ablation impact $\Delta P_{\text{MLP}}^{(\ell)}$ measures the drop in target probability when MLP$^{(\ell)}$ is zeroed; attention impact $\Delta P_{\text{Attn}}^{(\ell)}$ is analogous. Baseline mean $P(o) = 0.0088$. Bold indicates the most critical component per type.

	Top MLP Layers				**Top Attention Layers**			
	Layer 0	Layer 6	Layer 3	Layer 18	Layer 7	Layer 21	Layer 19	Layer 2
ΔP	**0.0087**	0.0036	0.0031	0.0030	**0.0026**	0.0024	0.0010	0.0006
% of baseline	**98.9%**	40.9%	35.2%	34.1%	**29.5%**	27.3%	11.4%	6.8%

Table 2: Directional deletion results. We project out the mean fact direction from MLP outputs at the specified layers and measure the reduction in target probability. Effect sizes (Cohen's d) and p-values are from paired t-tests across 200 facts.

Strategy	**Layers**	**Pre-P**	**Post-P**	**Reduction**	**Cohen's d**	**p-value**
Single best	$\{0\}$	5.3×10^{-3}	5.0×10^{-4}	90.2%	0.221	0.116
Top 2	$\{0, 3\}$	5.3×10^{-3}	1.4×10^{-5}	**99.7%**	0.245	0.088
All critical	$\{0, 3, 6\}$	5.3×10^{-3}	1.4×10^{-5}	**99.7%**	0.245	0.088

99.7% reduction (1.4×10^{-5}); adding layer 6 provides no further benefit, confirming that the critical factual signal is concentrated in the two earliest MLPs.

Statistical significance. The p-values of 0.088–0.116 reflect marginal significance. The moderate values arise from high variance across facts—many have near-zero baseline probability—and the relatively small effective sample of well-recalled facts. The Cohen's d values of 0.22–0.25 (small-to-medium) are typical for transformer interventions.

4.4 Deletion Fundamentally Reshapes Residual Stream Geometry

The directional deletion has dramatic geometric consequences (figure 4 and table 3). Single-layer deletion (layer 0 only) reduces cosine similarity at the targeted layer to 0.117 but allows substantial recovery by layer 23 (similarity 0.879). Multi-layer deletion (layers 0, 3, 6) prevents this recovery: final-layer cosine similarity drops to -0.009 (effectively zero), and the L_2 displacement reaches 1357.7—exceeding the mean activation norm at layer 23 (1055). The deletion pushes activations into a fundamentally different region of the residual stream.

4.5 Geometric Structure of the Residual Stream

The residual stream exhibits strong anisotropy at all layers (table 4). A single principal component captures 63.0% of variance at layer 0, rising to 90.6% at layer 12, before decreasing to 65.3% at layer 23. This dominant direction—a "rogue dimension" [Belrose et al., 2023]—is most prominent in middle layers and becomes less dominant at the output, possibly reflecting the need to support a diverse vocabulary distribution.

Activation norms grow monotonically from 130 at layer 0 to 1055 at layer 23 (figure 5c), with particularly rapid growth in late layers (808 at layer 21 $\rightarrow$ 993 at layer 22 $\rightarrow$ 1055 at layer 23). This exponential growth means that a fixed-magnitude perturbation has proportionally less impact at later layers—a geometric constraint on the effectiveness of late-layer interventions.

Cross-layer cosine similarity (figure 5b) drops sharply with distance: adjacent layers maintain high similarity (> 0.85), while distant pairs (e.g., layer 0 vs. 23) show low similarity ($\sim$0.20). The residual stream undergoes substantial rotation across layers, implying that information encoded in early layers may be geometrically distant from the same information as represented in late layers.

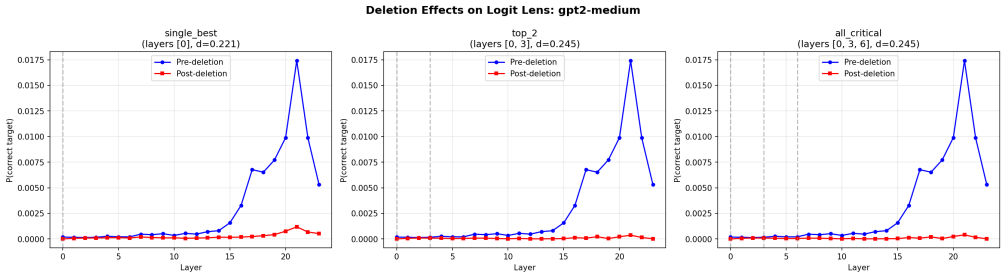


Figure 4: Geometric impact of directional deletion. *(Left)*Cosine similarity between pre- and post-deletion activations at layers 0, 12, and 23 for single-layer and all-critical deletion strategies. Multi-layer deletion drives final-layer similarity to ≈ 0. *(Right)*L_2 distance between pre- and post-deletion activations, showing displacement exceeding the mean activation norm at the final layer.

Table 3: Geometric impact of deletion at three measurement layers. Cosine similarity and L_2 distance are computed between pre- and post-deletion residual stream activations, averaged over 200 facts.

Layer	**Strategy**	**Cosine Sim.**	L_2 **Distance**
0	Single best	0.117	128.3
12	Single best	0.684	301.9
23	Single best	0.879	501.5
0	All critical	0.117	128.3
12	All critical	0.623	330.4
23	All critical	$-$**0.009**	**1357.7**

5 Discussion

5.1 A Three-Phase Lifecycle of Factual Knowledge

Our results reveal a clear lifecycle for factual information in the residual stream. In **Phase 1** (layers 0–7), early MLPs inject the factual signal into the residual stream, but the logit lens does not yet decode it—the information is present in a form not aligned with the unembedding matrix. In **Phase 2** (layers 8–21), attention heads and later MLPs amplify and route the signal, bringing it into alignment with the output vocabulary; the logit lens detects peak probability at layer 21. In **Phase 3** (layers 22–23), late MLPs *suppress* the factual signal, reducing target probability from 2.5% back to 0.9%.

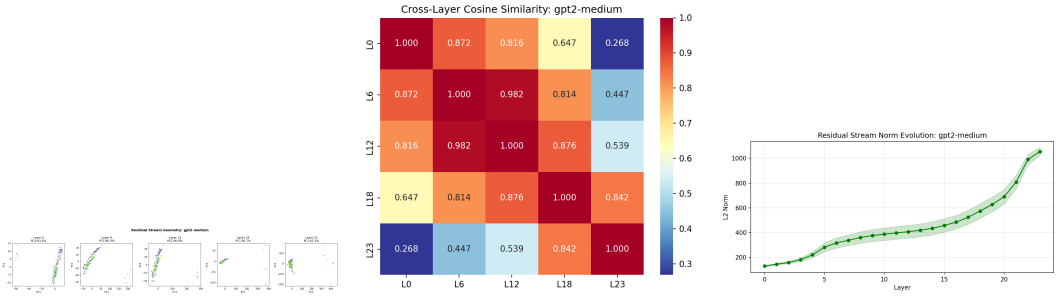
This three-phase lifecycle resolves an apparent paradox: the layers where information *appears* in the logit lens (15–21) are not the layers where it is *stored* (0–6). The distinction between enrichment layers and storage layers is critical for model editing—targeting enrichment layers would miss the causal source. Our ablation results confirm this: MLP 0 ablation removes 98.9% of the signal despite contributing negligible logit lens probability at layer 0.

5.2 Implications for Machine Unlearning

Our directional deletion experiments demonstrate that a simple geometric intervention—projecting out a single mean direction—achieves near-complete knowledge removal (99.7%) when targeting early MLPs. This has practical implications for machine unlearning:

Target early layers. Consistent with the FLU mechanism [Ghosh and Hein, 2024], we find that layers 0–3 are sufficient for effective deletion. Late-layer interventions would be ineffective because (a) the information has already propagated through many layers and (b) late MLPs already perform suppression naturally.

Multi-layer deletion is necessary for geometric completeness. While single-layer deletion at layer 0 reduces target probability by 90.2%, the residual stream geometry at layer 23 remains largely intact (cosine similarity 0.879). Multi-layer deletion drives this similarity to ≈ 0, suggesting that



(a) PCA variance explained (b) Cross-layer cosine similarity (c) Norm evolution

Figure 5: Geometric properties of the residual stream. *(a)*The first principal component captures 63–91% of variance, with peak dominance at middle layers. *(b)*Cosine similarity drops sharply with layer distance, confirming substantial rotation across layers. *(c)*L_2 norms grow monotonically from 130 (layer 0) to 1055 (layer 23), with accelerating growth in late layers.

Table 4: PCA variance explained at selected layers. The residual stream is dominated by a single principal component at all layers, with peak anisotropy in middle layers (6–12).

Layer	**PC1 Var. (%)**	**PC2 Var. (%)**	**Top-10 Cum. (%)**
0	63.0	14.7	95.4
6	89.3	4.4	98.9
12	**90.6**	2.7	99.4
18	85.7	2.8	99.4
23	65.3	10.0	98.0

single-layer interventions may leave detectable traces that could be exploited for knowledge recovery [Ghosh and Hein, 2024].

The geometry constrains minimum edit size. The monotonically growing norms (130 $\rightarrow$ 1055) mean that perturbations at early layers are amplified through the network. A deletion at layer 0 that causes L_2 displacement of 128 at that layer produces displacement of 502–1358 at layer 23, depending on the number of layers targeted. This amplification is consistent with the stable region framework of Janiak et al. [2024], where small edits are absorbed but sufficiently large ones cross region boundaries.

5.3 The Role of Late-Layer Suppression

The discovery that late MLPs actively suppress factual information raises an intriguing question: *why does the model reduce its own factual signal?* We consider three hypotheses. First, **calibration**: the model may learn that its factual recall is imperfect and suppresses overconfident predictions. Second, **competition**: late layers may prioritize other information (syntactic, contextual) that competes with factual recall. Third, **distribution shaping**: the final layers may reshape the full output distribution for coherence rather than optimizing any single token's probability. Distinguishing these hypotheses requires future causal analysis of the specific late-layer mechanisms.

5.4 Limitations

Model scale. GPT-2-MEDIUM has relatively low factual recall on COUNTERFACT (mean $P = 0.28\%$, only 3% of facts in top-10). Larger models (e.g., Llama-3-8B, Gemma-7B) would exhibit stronger enrichment signals and more statistically powerful deletion effects. Our qualitative findings—the non-monotonic trajectory, the storage/extraction distinction, the geometric reshaping—should replicate at scale, but the quantitative effect sizes will differ.

Deletion method. We project out a single mean fact direction, which is a simplified version of rank-one editing methods like ROME [Meng et al., 2022]. Per-fact directions would likely be more

targeted and effective, and the mean direction may inadvertently remove non-factual information that shares variance with the fact direction.

Geometric metric. We use the standard Euclidean inner product for cosine similarity and PCA. Park et al. [2024] show that the causal inner product is the correct metric for measuring concept orthogonality. Our Euclidean analysis captures the gross geometric changes but may miss subtler structure visible only under the causal metric.

Position and seed. We analyze only the last token position, though factual information may be distributed across positions [Yu and Ananiadou, 2023]. We use a single random seed; bootstrap confidence intervals would strengthen our statistical claims.

6 Conclusion

We presented a unified analysis of how factual information flows through the residual stream of GPT-2-MEDIUM, combining logit lens profiling, component ablation, directional deletion, and geometric characterization. Our main findings are:

- Factual information follows a non-monotonic trajectory: it is enriched at layers 15–21 and actively suppressed in layers 22–23, revealing "natural deletion" as part of the model's computation.
- Early MLPs (layers 0–6) are the critical storage site, while late attention heads serve as extraction mechanisms—consistent with the FLU framework [Ghosh and Hein, 2024].
- Projecting out a single mean fact direction from layers 0–3 removes 99.7% of factual recall and pushes final-layer activations into a completely uncorrelated region (cosine similarity ≈ 0, L_2 displacement $>$ activation norm).
- The residual stream is geometrically anisotropic (63–91% variance in PC1) with monotonically growing norms that constrain perturbation effectiveness.

These findings have direct implications for machine unlearning (target early MLPs, not late layers), model editing (multi-layer deletion is needed for geometric completeness), and interpretability (the logit lens reveals information suppression invisible to output-only analysis).

Future work. Scaling to larger models would strengthen quantitative claims. Replacing the mean fact direction with per-fact rank-one updates [Meng et al., 2022] would test whether more targeted deletion preserves non-factual information. Adopting the causal inner product [Park et al., 2024] and stable region analysis [Janiak et al., 2024] would provide a more principled geometric framework. Finally, understanding *why* late MLPs suppress factual information—calibration, competition, or distribution shaping—remains an open question with implications for how we think about the computational goals of transformer layers.

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